

Detecting Marine Heatwaves in the North Atlantic: Threshold-Based Analysis of Sea Surface Temperature Variability and Climate Drivers

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Abstract

This study investigates the frequency, duration, and intensity of marine heatwaves (MHWs) in the North Atlantic and explores their relationship with long-term anthropogenic and natural climate drivers. Using the Hobday et al. (2016) framework implemented through the heatwaveR package in R sea surface temperature (SST) time series were analyzed to identify MHW events. Preliminary results demonstrate an increase in MHW frequency and intensity in recent decades, aligning with elevated atmospheric CO₂ levels and warm phases of the Atlantic Multidecadal Oscillation (AMO).

Introduction

A marine heatwave is defined as a period of at least 5 consecutive days when SST is higher than the 90th percentile for that time of year (Hobday et al., 2016). They are influenced by atmosphere-ocean heat flux, horizontal and vertical advection, and global warming. Higher local atmospheric pressure systems, low surface level relative humidity, and warmer surface air temperatures drive higher SSTs. Changes in winds such as the weakening Pacific trade winds during El Niño allows for the eastern Pacific to warm more than usual near the equator (Holbrook et al., 2019).

The AMO is a proposed model of SST variability in the North Atlantic. While its drivers are unclear, the pattern shows the Atlantic is in a warming stage of an average of 0.10°C which could be impacting MHW frequency and intensity (Clement et al., 2015). Anthropogenic climate change is also looked at as a driver of mean ocean warming that would impact MHW frequency, intensity, and duration (Holbrook et al., 2019).

Methodology

After verifying the algorithm with a synthetic dataset, the workflow was applied to real SST data from the Northwest Atlantic included with the heatwaveR package. This dataset provided daily SST observations. The baseline climatology for the Northwest Atlantic was computed over 1982–2024 and MHW events were detected.

Results

- There was an increase the number of MHWs beginning around 2009.
- There were 124 MHW events from 1982-2024
- The seasonal climatology shows more variable SSTs in the spring and fall, and less variable in the winter and summer.
- The hottest MHW events tend to occur June-October

Figure 1. A) Time series from 1982-2023 showing SST with MHW events marked in yellow. A & B) Blue line represents the average SST. Red dashed line represents the 90th percentile extreme warm temperature that needs to be reached for an MHW. B) Mean and 90th percentile SST over a given year. Narrower gaps show times when the threshold can be more easily reached. C) Number of MHWs per year from 1982-2023 (Hobday et al., 2016).

Discussion

The increase in frequency, intensity, and duration of MHWs from 1982-2024 (Fig. 1a) aligns with broad patterns of ocean warming correlated with an increase in greenhouse gas emissions (Oliver et al., 2021). Long-term ocean warming elevates the baseline SST, increasing the probability that relatively small anomalies exceed the 90th-percentile threshold defined by Hobday et al. (2016).

The warm phase of the AMO likely elevated the background temperature state, making exceedance of the 90th-percentile threshold more common during recent decades. The clustering of recent events and aligns with modeling studies showing strong interactions among ocean circulation, radiative forcing, and SST variability (Clement et al., 2015).

Additionally, seasonal patterns can be seen in the assessment as the hotter events are occurring Jun-Oct (Fig. 3) as the ocean warms throughout the summer and stays warm into the fall due to the specific heat capacity of water (Holbrook et al., 2019).

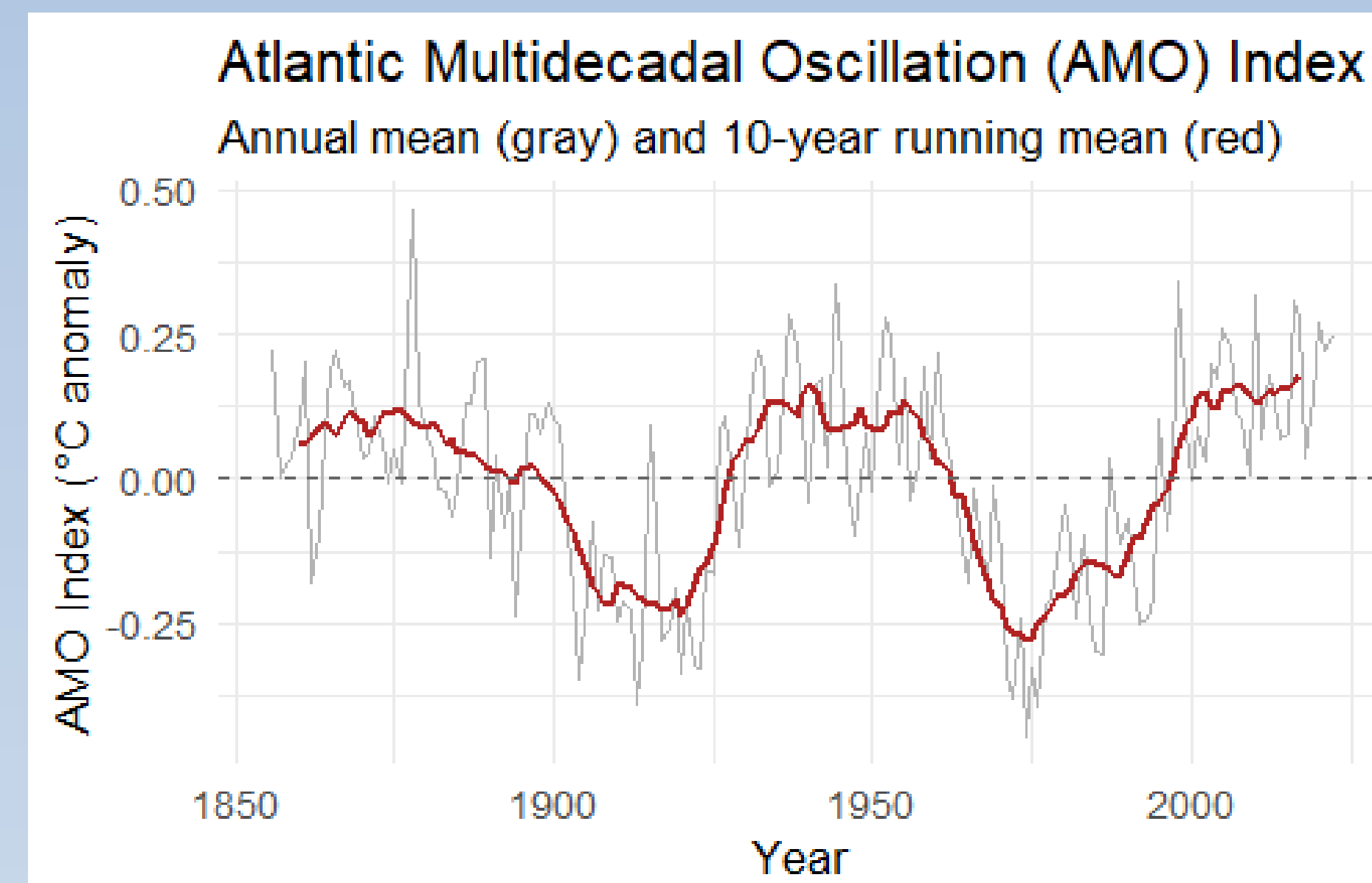


Figure 2. Atlantic Multidecadal Oscillation Index from 1856-2024. The gray line shows annual SST anomalies. The red line shows the 10-year running mean to highlight multidecadal variability. NOAA Kaplan SST dataset.

Marine Heatwaves — Detection and Summary

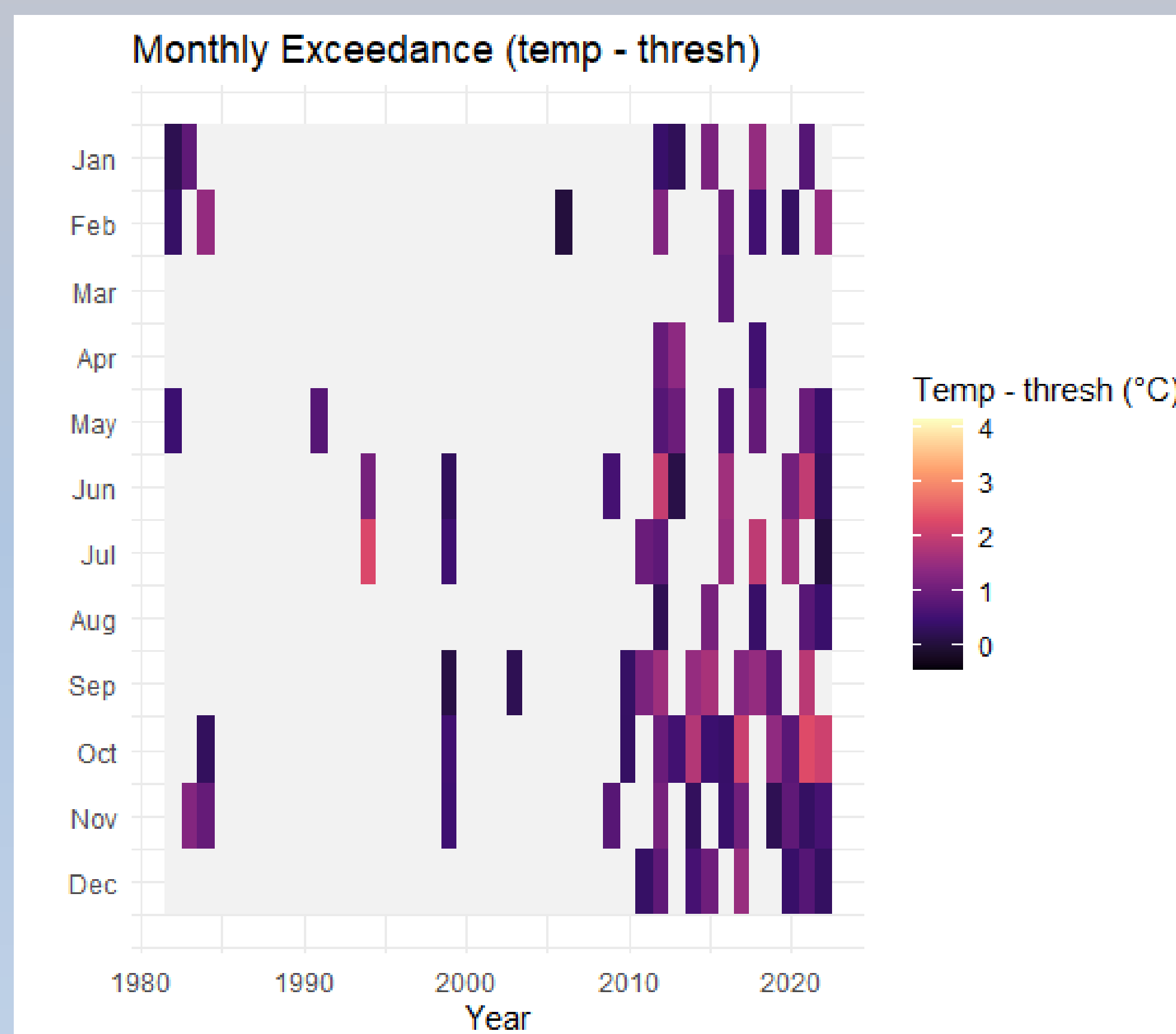
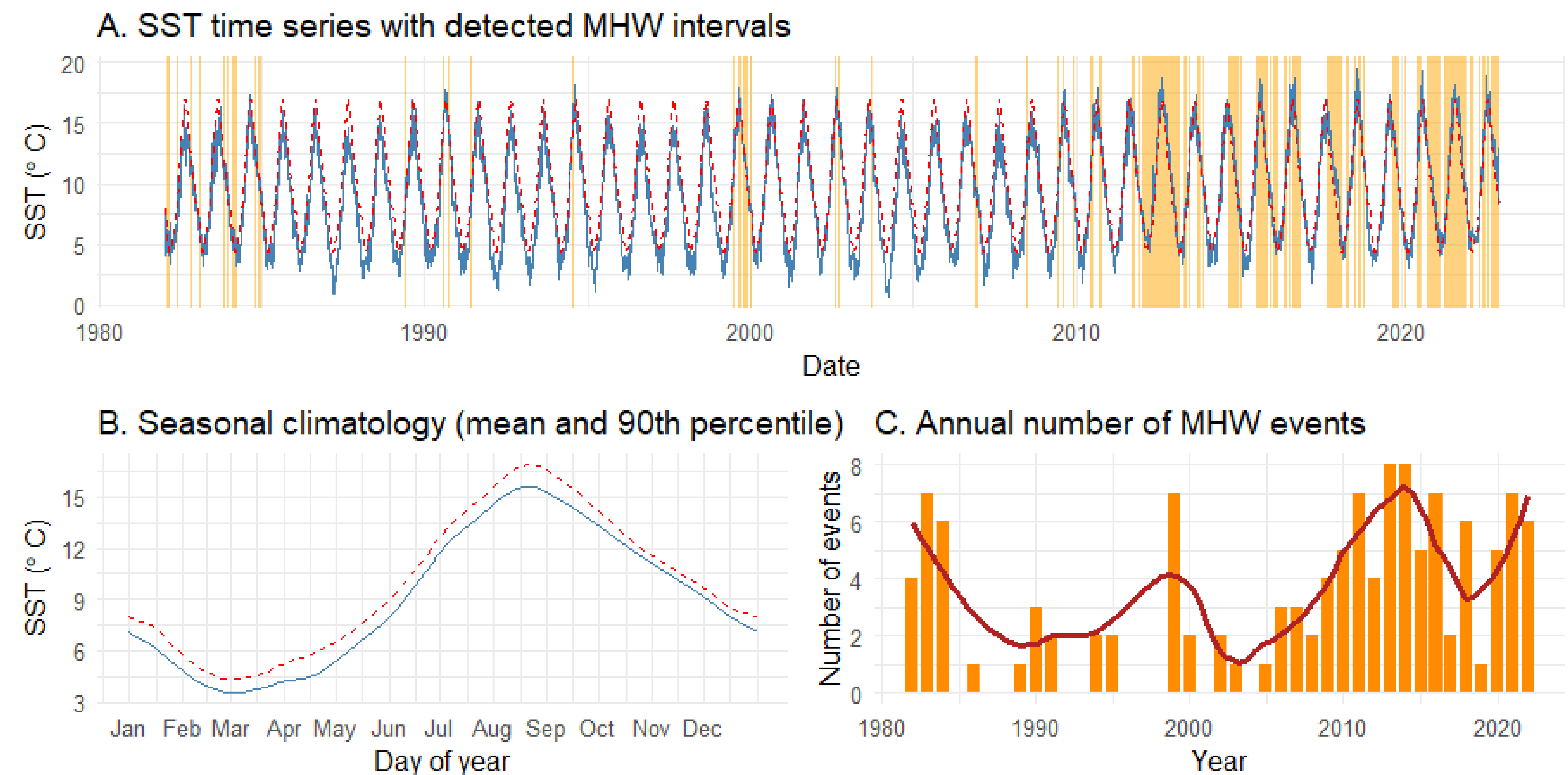


Figure 3. Color scale shows the degrees in Celsius above the 90th percentile SST threshold that an MHW mean SST was. Visualized for each month from 1982-2024 (Hobday et al., 2016).

Conclusions / Significance

Marine heatwaves in the Northwest Atlantic have increased in frequency and intensity in recent decades, with a notable rise after 2009. This trend reflects the combined influence of long-term anthropogenic warming—elevating baseline SSTs—and natural climate variability such as the AMO, which can further push temperatures above the extreme-warm threshold (Hobday et al., 2016; Clement et al., 2015; Holbrook et al., 2019).

MHWs are a relatively new area of scientific interest, with the first published use of the term being in 2011 (Oliver et al., 2021). They are an ecologically and economically important area of research and some of the fastest warming is occurring in the Northwest Atlantic, impacting important fisheries and ecosystems. In 2012, an MHW associated with Hurricane Sandy caused immediate shifts in the geographic ranges and seasonal cycles of species. Seasonal patterns indicate that the most intense events occur from June–October, when warm atmospheric conditions and ocean heat retention promote multi-day extremes. Stronger hurricanes could form as a result of an MHW (Mills et al., 2013).

As the Northwest Atlantic continues to warm faster than the global ocean, MHWs are expected to become more common, longer lasting, and more impactful. Continued monitoring and linking of these events to climate drivers are essential for anticipating ecosystem responses and informing adaptation strategies for coastal communities and fisheries.

Citations

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